PAPER 1 / WORKBOOK REVIEW

Answers to the 30 workbook questions

Questions 1–30 in workbook order. Section and table references follow the November manuscript and its supplements.

Market model and identification

1. State-space realization

Write R = diag(r_i), r_i = 1 - eta_i, and q for trading intensities. The price updates as x_t = a x_(t-1) + kappa q^T y_(t-1) + kappa v_t. Beliefs update as y_t = R y_(t-1) + eta x_t. The state matrix has blocks [a, kappa q^T; a eta, R + kappa eta q^T]. The input vector is kappa(1, eta)^T and the output selects the price. In the pre-update convention the output matrices are CA and CB = kappa. With s = z^(-1), H = kappa/[1 - as - kappa s sum_i q_i eta_i/(1-r_i s)]. The workbook formula follows from the separate negative-feedback equation p = kappa H_plant(u-qp).

2. Poles, zeros, and stability

For the workbook plant N/D = sum_i w_i z/(z-eta_i), the closed-loop transfer kappa N/(D+kappa qN) takes value 1/q at each nondegenerate eta_i. Those rates are zeros of sensitivity D/(D+kappa qN). For positive weights the plant is decreasing between its poles, giving one closed-loop pole in (0,eta_1) and one per gap. In the manuscript the numerator is kappa z product_i(z-r_i), giving memory zeros at nondegenerate r_i. Positive-cohort stability is a + kappa sum_i q_i < 1.

3. Pole matching and identification

Normalized weights, rates, and two primitives give 2K+1 free parameters. A rank-K pole map has a local level set of dimension K+1. This establishes pole matching only. Under the manuscript's single-innovation minimum-phase assumptions, the spectrum also identifies the transfer function up to scale, and known primitives identify the merged measure.

4. Merged measures and moments

For common zero initial conditions, equality for all inputs is equivalent to equality of impulse responses. With M(s)=sum_j m_j s^j, invert H to obtain M=(1-as-kappa/H)/(kappa s). Equality of taps h_0 through h_(2K) gives equality of moments m_0 through m_(2K-1). The difference of two K-atomic measures has at most 2K nodes, and the Vandermonde matrix forces every signed weight to be zero.

5. Stock and flow roots

If S_t=rho S_(t-1)+u_t and u_t=r u_(t-1)+b R_t, substitution gives S_t-(r+rho)S_(t-1)+r rho S_(t-2)=b R_t. The roots are r and rho, both potentially stable. Changes in holdings are u_t-(1-rho)S_(t-1). The weekly two-root test passed in 2 of 8 markets. See Supplement S18.

6. Plug-in estimation

The finite tap-to-moment map is smooth when kappa stays away from zero. Baseline demeaning removes constant p*. Root-T errors in taps and primitives preserve the moment error order and the subsequent spacing-dependent rate. From h_0=kappa and h_1=a kappa+kappa^2 m_0 alone, a is confounded with m_0. See Corollary 5.3.

Minimax bounds and Prony recovery

7. Moment-matched hard pairs

Take 2K equally spaced nodes with alternating weights proportional to (-1)^j times binomial(2K-1,j). Splitting by sign produces two K-atom measures. Finite differences annihilate degrees 0 through 2K-2, and the first unmatched moment is order Delta^(2K-1). The one-lag feedback makes taps agree through 2K-1.

8. KL divergence and total variation

For N independent Gaussian resets with covariance sigma^2 I, KL=N times the squared response gap divided by 2 sigma^2. A Taylor expansion on a compact stable rate interval gives a square-summable kernel remainder of order Delta^(2K-1). Uniform stability transfers it to the response. If the squared gap is at most C_K Delta^(4K-2), Pinsker bounds TV by sqrt(C_K N/(4 sigma^2)) Delta^(2K-1).

9. Le Cam sample requirement

Worst-case error at most alpha requires TV at least 1-2alpha. Therefore N >= 4(1-2alpha)^2 sigma^2 Delta^(-(4K-2))/C_K. Any estimator accurate enough to separate the alternatives yields a test.

10. Spacing multipliers

Shrinking the gap from 0.10 to 0.01 multiplies the experiment order by 10^6 for two cohorts and 10^10 for three, holding other inputs fixed.

11. Prony-Hankel recovery

Deconvolve price taps to obtain 2K moments. Build the K-by-K matrix L_ij=m_(i+j) and solve Lc=-(m_K,...,m_(2K-1))^T. The roots of the monic annihilating polynomial give rates. Solve a Vandermonde system for weights, then use eta_i=1-r_i and q_i=w_i/eta_i.

12. Hankel conditioning

A 2K-by-2K Hankel matrix from K atoms has rank at most K. For the K-by-K matrix L=V diag(w_i)V^T, positive bounded weights and uniformly separated rescaled nodes give smallest Vandermonde singular value of order Delta^(K-1) and smallest Hankel singular value of order Delta^(2K-2).

13. Matching upper bound

For tap noise e=sigma/sqrt(N), the largest centered, rescaled moment error is O_p(e Delta^(-(2K-1))). The inverse rescaled moment map is locally uniform. Weight errors have that order and original-rate errors gain one Delta. At weight tolerance epsilon and rate tolerance epsilon Delta, N is of order sigma^2 epsilon^(-2) Delta^(-(4K-2)). See Theorem 5.2.

14. ESPRIT and matrix pencil

The hard pair limits every estimator with the same observations. ESPRIT and matrix pencil can differ in finite-sample behavior and numerical conditioning.

Probing, passive data, and disclosure

15. Positive round-trip cost

Positive decreasing taps alone are insufficient. The Toeplitz matrix for (1,0.9,0.1) has determinant -0.468. Positive mixtures of decaying exponentials instead have positive spectral densities, giving a strictly positive quadratic cost for nonzero trades.

16. Probing energy

A deterministic probe has mean gap (h_1-h_0)*u. If the transfer gap is bounded by L epsilon Delta^(2K-1), KL is bounded by L^2 epsilon^2 Delta^(4K-2) times squared probe energy divided by 2sigma^2. Le Cam yields the reciprocal energy order.

17. Funded capital barrier

The symmetric known pair has transfer gap O(delta^2). Fixed testing error therefore needs energy of order at least sigma^2 delta^(-4). Coercive quadratic funded execution converts that to a capital lower bound under the stated contract. Halving the separation gives a factor of 16. See Lemma S24.4.

18. Annual dollar calibration

Table S10 gives $14.74 million and $356.44 million annually per market at labeled gap Delta eta=0.10, using Roll and one-second noise. Weekly loss figures of about $283,488 and $6,854,612 are multiplied by 52.

19. Passive spectral divergence

The Whittle objective sums log f plus periodogram divided by f. Gaussian KL rate integrates f_0/f_1-1-log(f_0/f_1) with factor 1/(4pi). Uniform spectral ratio perturbation O(Delta^(2K-1)) yields divergence O(Delta^(4K-2)) by a quadratic expansion. A matching order needs a nonzero leading spectral change. See Theorem 5.5.

20. Bitcoin resampling

Multinomial counts with equal probabilities reproduce independent draws with replacement from the empirical shock pool. The reported slopes were -6.01 and -9.81 for synthetic two- and three-cohort signals with real Bitcoin noise. See S16 and Table S23.

21. General memory recovery

Moment matching to degree L on [0,rho], rho<1, can keep measures separated by order 1/L while shrinking the response gap geometrically. Choose L proportional to log(N/sigma^2) for the testing bound. Polynomial approximation of Lipschitz functions gives the matching recovery order 1/log(N/sigma^2). See S17.2.

22. Tagged disclosure

Tagged taps f_i,0=w_i h_0 and f_i,1=w_i(h_1+r_i h_0) identify the weight and rate by division. With h_0>0 and weights bounded away from zero, root-N accuracy is independent of spacing. Fixed weight tolerance is constant in spacing and rate tolerance proportional to the gap costs order Delta^(-2).

23. Supervisory choice

With costs C_rep and A Delta^(-p) alone, reporting is cheaper below (A/C_rep)^(1/p). An intermediate band requires an additional option or a gap-dependent value of attribution.

Empirical findings and policy

24. Eight-market CFTC study

CME Bitcoin, CME Ether, E-mini S&P 500, Treasury notes, euro, yen, WTI and natural gas form the primary sample. The speculative classes are leveraged funds or managed money. Comparators are asset managers or producers/merchants. Tagged intervals were 9.1-71.4 times narrower. H1 passed 8/8, H2 failed at 4/8, and H3 passed 6/8. See Table S25.

25. Twelve-market replication

Gold, silver, copper, corn, soybeans, wheat, cattle, coffee, sugar, cotton, Nasdaq-100 and the British pound are the new markets. Tagged widths were 0.014-0.042 versus 0.85-0.96 for prices. H1 passed 12/12, H2 passed 11/12, and H3 failed at 3/12. Table S26 gives coffee about 4,735 times the sample and a 20-market median of 1,353.

26. Empirical scorecard

Reported passes or holds include CFTC H1 8/8 and H3 6/8, noise K=2 and K=3, two Binance fresh-market checks 30/30, horizon-matched 29/29, tagging advantage 22/29, state tracking, replication H1 12/12 and H2 11/12, and N3 shrinkage 30/30. Failures include CFTC H2 4/8, weekly flow 3/8, two-root 2/8, Binance combined response 5/30, 105-day 10/30, 58-day shrinkage 20/29, return forecast -1.09 points, replication H3 3/12, October N2 1/30, and October N3 5/30. Binance stable learning memory was unsupported. Table S29 also lists N3 sign stability as a failure at 21/30 versus 24 required.

27. Within-class cancellation

Opposing subgroup signals cancel when their weighted magnitudes nearly match. Distinct exponential rates cannot cancel exactly for every common input with nonzero weights. Failed aggregate tests alone do not identify the mechanism.

28. Spectral blind spot

For triangular A with diagonal eigenvalues lambda_1 and lambda_2 and coupling gamma, the upper off-diagonal of A^t is gamma(lambda_1^t-lambda_2^t)/(lambda_1-lambda_2). At fixed eigenvalues peak gain grows linearly in gamma. Stationary price variance equals s_1/(1-lambda_1^2) plus gamma^2 s_2 times the sum of squared divided differences. Taking s_2 proportional to gamma^(-2) holds stationary variance fixed for positive gamma.

29. Pseudospectra and Kreiss bound

The epsilon-pseudospectrum contains eigenvalues and points where the resolvent norm exceeds 1/epsilon. The Kreiss constant is sup over

30. Inventory decisions and targeted halts

For independent tap noise across N resets the finite-horizon sum G has standard error sigma sqrt(H/N). Nonnegative taps and bounded shocks let an upper confidence bound on G size exposure. A targeted halt asks whether removing one cohort crosses a stability threshold, and hard pairs inherit the testing lower bound. See Theorem 0.3 and Proposition 6.4.